Technical Supplement

Reproducibility notes for A Deterministic Evidence Gate for Pharmacogenomic LLM Alerts: Synthetic Stress-Testing with Ablation

Safety Boundary

Synthetic-only research artifact. No real patient data, protected health information, diagnosis, or treatment recommendation. The deterministic monitor evaluates claim permission levels from structured annotations; text-only extraction is evaluated separately and remains below the oracle annotation ceiling.

Reproducibility Commands

- py -3.13 code\pruned_evidence_gate.py
- py -3.13 code\llm_self_evaluation_baseline.py
- py -3.13 code\text_only_extraction.py
- py -3.13 code\analyze_text_extraction_outputs.py
- py -3.13 code\heldout_case_authoring.py
- py -3.13 code\generate_paper_assets.py
- py -3.13 -m pytest code -q
- py -3.13 code\build_submission_package.py

Main Synthetic Case Bank

The final case matrix contains 33 author-designed synthetic cases: 10 bounded aligned alerts and 23 designed overclaim archetypes. The 23/10 split is stress-test coverage, not a measured clinical base rate.

Metric	Value
Case count	33
Author-designed overclaim cases	23
Bounded aligned alerts	10
Full monitor overclaims allowed unchanged	0/23
Bounded alerts preserved	10/10
Inappropriate denials	0/10
Action conformance	33/33
Primary-check conformance	31/33
Precedence-sensitive cases	PGX19; PGX24

Ablation and Comparator Snapshot

Analysis	Result
Disable source support	2/23 overclaims pass unchanged; 7 action changes; 12 primary-check changes.
Disable population fit	1/23 overclaim passes unchanged; 5 action changes; 5 primary-check changes.
Disable claim strength	6/23 overclaims pass unchanged; 6 action changes; 6 primary-check changes.
Ungated allow-all	23/23 overclaims pass unchanged.
Claim-strength-only	3/23 overclaims pass unchanged: PGX31, PGX32, PGX33.
Full three-check monitor	0/23 overclaims pass unchanged.

Structured-Label Translation Arms

Arm	Scope	Result
Arm A GPT-5.6-terra	Full 33-case bank with source-support labels	23/23 overclaims routed; 10/10 bounded alerts preserved; 31/33 action agreement; 25/33 primary-check agreement.
Arm A Grok-4.5	Full 33-case bank with source-support labels	23/23 overclaims routed; 10/10 bounded alerts preserved; 31/33 action agreement; 32/33 primary-check agreement.
Arm B GPT-5.6-terra	PGX31-PGX33; source-support label withheld	Missed PGX31; matched PGX32 and PGX33.
Arm B Grok-4.5	PGX31-PGX33; source-support label withheld	Matched PGX31 and PGX32; PGX33 action differed while primary check matched.
Uniform-NARROW sanity row	Synthetic, no API call	23/23 overclaims routed; 0/10 bounded alerts preserved; 33/33 narrowed.

Text-Only Extraction

Provider / condition	Field accuracy	Downstream action	Oracle gap
Regime 1: GPT-5.6-terra with citation	78/132	21/33	4/10 bounded preserved; 6/10 bounded losses, all NARROW_CLAIM; inappropriate denials 0.
Regime 1: Grok-4.5 with citation	86/132	21/33	9/10 bounded preserved; 1/10 bounded loss, NARROW_CLAIM; inappropriate denials 0.
Regime 2: GPT-5.6-terra no citation	73/132	11/33	0/10 bounded preserved; 10/10 bounded losses, all ABSTAIN_SOURCE_SUPPORT; inappropriate denials 10.
Regime 2: Grok-4.5 no citation	70/132	12/33	2/10 bounded preserved; 8/10 bounded losses, all ABSTAIN_SOURCE_SUPPORT; inappropriate denials 8.

This is the main bottleneck result: the deterministic monitor performs as specified under oracle annotations, but LLM extraction from text and citation fields loses annotation accuracy and changes downstream actions. With citation fields, every lost bounded alert was narrowed rather than denied or abstained: 6/6 for GPT-5.6-terra and 1/1 for Grok-4.5. With citations withheld, every lost bounded alert became ABSTAIN_SOURCE_SUPPORT: 10/10 for GPT-5.6-terra and 8/8 for Grok-4.5. Both conditions routed 23/23 overclaims and allowed 0 overclaims unchanged, so bounded-alert preservation is the metric that separates useful review from a degenerate abstention pattern.

Directional Error and Field PRF

Analysis	Result
Error direction, GPT with citation	45 conservative errors vs. 9 permissive errors.
Error direction, Grok with citation	42 conservative errors vs. 4 permissive errors.
Error direction, GPT no citation	43 conservative errors vs. 16 permissive errors.
Error direction, Grok no citation	56 conservative errors vs. 6 permissive errors.
Macro-F1, GPT with citation	Citation 0.7479; population 0.4369; endpoint 0.7008; actionability 0.3872.
Macro-F1, Grok with citation	Citation 0.6516; population 0.5974; endpoint 0.6590; actionability 0.4982.

Full macro-precision, macro-recall, and macro-F1 for every field, model, and condition are written to results/text_only_extraction_field_prf.csv. Several enum cells contain one or two observations, so macro-F1 is indicative rather than precise.

Held-Out Worksheet

GPT-5.6-sol generated 12 draft scenarios for a future held-out worksheet. They remain unlabeled in this package. The CSV worksheet has blank label columns, and no held-out accuracy, independence claim, or generalization result is reported.

Traceability

Every number in the paper is mapped to a generated file in results/paper_number_trace.csv. Raw model payloads are retained under results/*raw.json. Regression tests assert that author-only annotation_note fields are not sent in any model payload.